Fiscal Year:	FY 2017	**Task Last Updated:**	FY 03/01/2017
PI Name:	Motil, Brian J. Ph.D.		
Project Title:	Modeling and Experimental Studies on Gas-Liquid Two-Phase Flow Through Packed Beds in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	brian.j.motil@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	216-433-6617
Organization Name:	NASA Glenn Research Center		
PI Address 1:	21000 Brookpark Road		
PI Address 2:	GRC-LTZ0		
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	10
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	03/01/2014	**End Date:**	05/31/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Motil, Brian J.	**Contact Phone:**	216-433-6617
Contact Email:	brian.j.motil@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed back to 5/31/2017 per NSSC information (Ed., 4/28/21) NOTE: Extended to 2/28/2018 per PI (Ed., 1/17/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Balakotaiah , Vemuri Ph.D. (University of Houston (grant NNX14AD28G))		
Grant/Contract No.:	Internal Project ; CoI grant NNX14AD28G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Experimental Studies on Gas-Liquid Two-Phase Flow Through Packed Beds in Microgravity," internal project--Principal Investigator Dr. Brian Motil ; CoInvestigator Vemuri Balakotaiah, Ph.D., grant NNX10AL37G. Period of performance was 4/12/2010-10/11/2013. A basic understanding of two-phase flows through porous media is of interest in many chemical and biological processing systems as well as many geophysical applications. NASA technologies include adiabatic two-phase flows or two-phase flows with heat transfer that are relevant to life support systems, fuel cells, in-situ resource utilization, heat pipes, materials processing, production of pharmaceutical grade water, and transport of nutrients in soils. A very important unit operation that involves gas-liquid flow through porous media is the fixed Packed Bed Reactor (PBR). Examples of current space systems in which a fixed PBR is used include the Volatile Removal Assembly (VRA), the Integrated Advanced Water Recovery System (AWRS), and IntraVenous Water GENeration system (IVGEN). NASA is currently operating systems with packed bed reactors for ground and flight-based testing with an incomplete understanding of how the reduced gravity environment affects the performance and reliability of these reactors. This is especially critical when gas-liquid flows are involved. The expected outcome of this research effort is to develop a set of guidelines and tools to enable engineers to reliably design and operate packed bed reactors for microgravity as well as the lunar and Martian environments. Existing approaches for predicting the flow regimes and pressure drop for gas-liquid flow through a PBR are valid only for 1 g, and are based on liquid (or gas) holdup within the bed. Initial microgravity tests conducted in the reduced gravity aircraft and drop tower clearly show that the liquid holdup is significantly influenced by gravity. Furthermore, most of these models have only been verified using the "trickle flow" regime which does not exist in microgravity. Basic hydrodynamic models developed specifically for reduced gravity have been partially validated within the constraints of aircraft and ground facilities. However, typical flow ranges used for real systems require up to several minutes to develop fully which is far beyond the time scales available for ground tests. In addition, instabilities and hysteresis effects that may exist require long testing durations to establish new operating conditions. Thus, testing on the International Space Station (ISS) is required for complete validation of the semi-empirical models. Flight experiments can provide critical hydrodynamic information for a project with a broader scope that includes reduced gravity aircraft and ground-based (1 g) experiments. The main objective of the overall project is to develop and validate macroscopic equations that can be used in partial and microgravity conditions to accurately predict flow pattern transitions; pressure drops; and mass transfer rates in chemical and biological reactors in which gas-liquid flows occur through randomly packed beds. The hydrodynamic investigations addressed in the Packed Bed Reactor Experiment (PBRE) focus on the transitions between flow regimes (e.g., bubbly-to-pulse flow transition) and the associated pressure gradients over the range of relevant operating regimes of the PBRs (e.g., liquid and gas flow rates, particle sizes, and fluid viscosities). These design tools provide a fundamental understanding of these phenomena in low gravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The PBRE provides a fundamental investigation of two phase flow through porous media in the absence of gravity. This enables a direct study of the other forces that impact the flow and reaction rates such as surface tension, inertia, and viscous forces. These new insights will impact reactor designs on Earth.
Task Progress:	The PBRE is projected to complete a revised test matrix on the ISS by February 10th, 2017. The test matrix was designed to study the fluid mechanics within a fixed bed reactor including: 1) minimum liquid flow to remove trapped bubbles; 2) steady state flows to determine pressure drop and flow distribution for a wide range of gas and liquid flows; 3) transient flows to determine hysteresis effects during startup or step changes in flow rates; and 4) low gas flows into a stationary liquid to observe viscous fingering phenomena. Two types of packing have been studied (glass beads and Teflon) to observe the effects of a wetting vs. non-wetting material. Testing began in June, 2016 and it was determined that the ISS/MSG could not provide the full range of gas flows as advertised. Gas flows up to 3 kg/hr were required, but only about 1.1 kg/hr is obtainable. The test matrix was revised to focus on the lower gas flow rates, which are the most critical. Approximately 25% of the highest gas flows were not completed. All flight data will not be available until 2-3 months after completion in February, but initial analysis indicates an unexpected gas-continuous phase at low liquid and gas flows and much higher pressure drops at high liquid flows than predicted. NOTE: project continues as Modeling and Analysis of Packed Bed Reactor Experimental Data from International Space Station (formerly Modeling and Experimental Studies on Gas-Liquid Two-Phase Flow Through Packed Beds in Microgravity), internal project with Dr. Brian Motil as PI and CoInvestigator Vemuri Balakotaiah, Ph.D., grant 80NSSC18K0813.
Bibliography Type:	Description: (Last Updated: 04/26/2024)
Articles in Peer-reviewed Journals	Salgi P, Balakotaiah V. "Experimentally-based constitutive relations for co-current gas-liquid flow in randomly packed beds." AIChE Journal. 2017 Feb;63(2):812-22. http://dx.doi.org/10.1002/aic.15377 , Feb-2017